TABLE 8.1 Basic integration formulas

1. $\int k\,dx = kx + C$ (any number k)
2. $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
3. $\int \frac{dx}{x} = \ln|x| + C$
4. $\int e^x\,dx = e^x + C$
5. $\int a^x\,dx = \frac{a^x}{\ln a} + C \quad (a > 0,\ a \neq 1)$
6. $\int \sin x\,dx = -\cos x + C$
7. $\int \cos x\,dx = \sin x + C$
8. $\int \sec^2 x\,dx = \tan x + C$
8. $\int \csc^2 x\,dx = -\cot x + C$
10. $\int \sec x \tan x\,dx = \sec x + C$
11. $\int \csc x \cot x\,dx = -\csc x + C$
12. $\int \tan x\,dx = \ln|\sec x| + C$
13. $\int \cot x\,dx = \ln|\sin x| + C$
14. $\int \sec x\,dx = \ln|\sec x + \tan x| + C$
15. $\int \csc x\,dx = -\ln|\csc x + \cot x| + C$
16. $\int \sinh x\,dx = \cosh x + C$
17. $\int \cosh x\,dx = \sinh x + C$
18. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$
19. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$
20. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{x}{a}\right| + C$
21. $\int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C \quad (a > 0)$
22. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C \quad (x > a > 0)$

Using right endpoints, the area of the region that lies under the graph $f(x) = \sqrt{\cos x},\ 0 \le x \le \frac{\pi}{2}$ is

(a) $\lim_{n\to\infty} \frac{\pi}{2n} \sum_{i=1}^{n} \sqrt{\cos\left(i\frac{\pi}{2n}\right)}$

(b) $\lim_{n\to\infty} \frac{\pi}{n} \sum_{i=1}^{n} \sqrt{\cos\left(i\frac{\pi}{n}\right)}$

(c) $\lim_{n\to\infty} \pi \sum_{i=1}^{n} \sqrt{\cos\left(i\frac{\pi}{n}\right)}$

(d) $\lim_{n\to\infty} \frac{\pi}{2n} \sum_{i=1}^{n} \sqrt{\cos\left(i\frac{\pi}{n}\right)}$

(e) $\lim_{n\to\infty} \frac{\pi}{2n} \sum_{i=1}^{n} \sqrt{\cos\left(\frac{i}{2n}\right)}$

Evaluate the limit

$$\lim_{n\to\infty} \sum_{i=1}^{n} \frac{5}{n} \sqrt[4]{1+\frac{i}{n}}$$

a) $4\,(\sqrt[4]{32}-1)$
b) $4\,(\sqrt[4]{8}-1)$
c) $2\,(\sqrt[4]{2}+1)$
d) $6\,(\sqrt[4]{32}+1)$
e) $8\,(\sqrt[4]{4}-1)$

Use three rectangles and mindpoints to estimate the area under the graph of $f(x) = \frac{x}{x+1}$ over the interval $[0, 3]$ 1/2 3/2 5/2

a) $\frac{173}{105}$
b) $\frac{122}{105}$
c) $\frac{18}{35}$
d) $\frac{19}{35}$
e) $\frac{31}{105}$

For $x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$, $\frac{d}{dx}\left[\int_{\tan x}^{5} (t - \tan^{-1} t)dt\right] =$

(a) $x \sec^2 x - \tan x \sec^2 x$

(b) $(5 - \tan^{-1} 5) - \tan x \sec^2 x$

(c) $\tan x \sec^2 x - x \sec^2 x$

(d) $(\tan x - \tan^{-1} x) \sec^2 x$

(e) $\tan x - x$

If $6 + \int_a^x \frac{f(t)}{t^2}\, dt = 2\sqrt{x}$, $x \geq 0$, then $f(x) + a =$

(a) $9 + \sqrt{x^3}$

(b) $9 + \sqrt[3]{x^2}$

(c) $6 + \sqrt{x^2}$

(d) $6 + \sqrt{x^3}$

(e) $9 + \sqrt{x}$

$\int_1^{64} \frac{1+\sqrt[3]{x}}{\sqrt{x}}\,dx =$

(a) $\frac{256}{5}$

(b) $\frac{216}{5}$

(c) $\frac{212}{5}$

(d) $\frac{144}{5}$

(e) $\frac{252}{5}$

$\int_0^{\frac{1}{\sqrt{3}}} \frac{x^2-1}{x^4-1}\,dx =$

(a) $\frac{\pi}{6}$

(b) $\frac{\pi}{2}$

(c) $\frac{\pi}{4}$

(d) $\frac{8}{9}$

(e) 1

$\int_0^3 8\sqrt{9-x^2}\,dx =$

Let b be a real number such that $0 < b < 2$. If the average value of $f(x) = \cos(2x)$ on the interval $[0, b]$ is **zero**, then $b =$

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{4}$

(d) $\frac{\pi}{5}$

(e) $\frac{\pi}{6}$

The average value of $f(x) = x \sin(x^2)$ over $[0, 10]$ is equal to

a) $\frac{1}{20}(1 - \cos 100)$

b) $\frac{1}{10}(\cos 10 - 1)$

c) $\frac{1}{10}(\cos 1 - 1)$

d) $\frac{1}{20}(\sin 100 - 1)$

e) $\frac{1}{10}(\sin 10 - 1)$

$$\int x^2\sqrt{2+x}\,dx =$$

(a) $\frac{2}{7}(x+2)^{7/2} - \frac{8}{5}(x+2)^{5/2} + \frac{8}{3}(x+2)^{3/2} + C$

(b) $\frac{x^3}{3} + \frac{2}{3}(2+x)^{3/2} + C$

(c) $\frac{1}{7}(2+x)^{3/2} - \frac{5}{8}(x+2)^{5/2} + C$

(d) $\frac{2}{7}(x)^{7/2} - \frac{5}{8}(x)^{5/2} + \frac{1}{3}(x)^{3/2} + C$

(e) $\frac{2}{5}(x+2)^{5/2} - 3(x+2)^{1/2} + 8(x+2)^{3/2} + C$

Find the area of the region bounded by the curves
$y = \ln x, \quad x = -1 - y^2, \quad y = 0, \quad y = 1$

The area enclosed between $y = \cos x$ and $y = 1 - \cos x$ over $[0, \pi]$ is equal to:

(a) $2\sqrt{3} + \frac{\pi}{3}$

(b) $\sqrt{3} + \frac{\pi}{3}$

(c) $\frac{2}{3}\pi + \sqrt{3}$

(d) $2\sqrt{3} + 3\pi$

(e) $2 + \frac{\pi}{3}$

The region bounded by the curve $y = x^2 + 1$ and the line $y = -x + 3$ is revolved about x-axis to generate the solid S. The volume of S is equal to:

(a) $\frac{117}{5}\pi$

(b) $\frac{131}{3}\pi$

(c) $\frac{110}{3}\pi$

(d) $\frac{109}{5}\pi$

(e) $\frac{100}{3}\pi$

The volume of the solid formed by revolving the region bounded by the graphs of $f(x) = 2 - x^2$ and $g(x) = 1$ about the line $y = 1$, is given by

(a) $\pi \int_{-1}^{1} (1 - x^2)^2 \, dx$

(b) $2\pi \int_{0}^{1} (1 + x^2)^2 \, dx$

(c) $\pi \int_{-1}^{1} (-1 - x^2)^2 \, dx$

(d) $\pi \int_{0}^{2} (-1 - x^2)^2 \, dx$

(e) None of other choices

The **volume** of the solid generated by revolving the region bounded by the curves $y^2 = -x$ and $x - y + 2 = 0$ about the line $y = 1$ is given by the integral

(a) $\int_{-2}^{1} 2\pi(y^3 - 3y + 2)\,dy$

(b) $\int_{-4}^{0} 2\pi(y^3 + y^2 + 3y - 2)\,dy$

(c) $\int_{-2}^{1} 2\pi(-y^3 + 3y - 4)\,dy$

(d) $\int_{-4}^{0} 2\pi(-y^3 + y^2 - 3y + 2)\,dy$

(e) $\int_{-2}^{1} 2\pi(y^3 + 2y^2 - 3y - 2)\,dy$

Using the method of cylindrical shell, the volume of the solid generated by revolvin about the x-axis the region bounded by $x = 1 + (y-2)^2$; $x = 2$, is equal to:

a) $\frac{16}{3}\pi$

b) $\frac{2}{3}\pi$

c) $\frac{4}{3}\pi + 2$

d) $5\pi + \frac{1}{4}$

e) $2\pi + 3$

(a) Find the value of $\int_2^{\infty} \frac{\mathrm{d}x}{x^2+x-2}$.

(b) $\int \frac{x^2}{\sqrt{49-x^2}}$ d

(c) $\int_0^{2\sqrt{3}-2} \frac{\mathrm{d}x}{x^2+4x+8}$

Problem 5 (5+5+5 points). In the following problem, all answers should be left in the form $\int_a^b h(x)\,\mathrm{d}x$; *do not evaluate the integrals.*

(a) If the part of the parabola $y = x^2$ that lies below $y = 1$ is revolved about the Y-axis, what is the volume of the resulting solid?

(b) Find the area of the region bounded by the graphs $y^2 = x$ and $y = x^3$.

(c) If the region given in (b) is revolved about the X-axis, find the volume of the resulting solid.

The volume of the solid having base the region bounded by the lines $y = 1 - \frac{x}{2}$ and $y = -1 + \frac{x}{2}$ and $x = 0$, where the cross sections perpendicular to the x-axis, are equilateral triangles is

(Hint: Area of an equilateral triangle of side s is $\frac{\sqrt{3}}{4}(s)^2$)

(a) $\frac{2}{3}\sqrt{3}$

(b) $3\sqrt{3}$

(c) $2\sqrt{3}$

(d) $\frac{3}{2}\sqrt{3}$

(e) $\sqrt{3}$

(c) $\int_0^1 \frac{5x\,dx}{(x+2)(x^2+1)}$.

1. Find $\int x(\ln x)^2 dx$.

8.4 Trigonometric Substitutions

Trigonometric substitutions occur when we replace the variable of integration by a trigonometric function. The most common substitutions are $x = a\tan\theta$, $x = a\sin\theta$, and $x = a\sec\theta$. These substitutions are effective in transforming integrals involving $\sqrt{a^2 + x^2}$, $\sqrt{a^2 - x^2}$, and $\sqrt{x^2 - a^2}$ into integrals we can evaluate directly since they come from the reference right triangles in Figure 8.2.

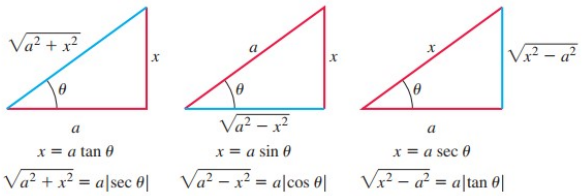


FIGURE 8.2 Reference triangles for the three basic substitutions identifying the sides labeled x and a for each substitution.

With $x = a\tan\theta$,

$$a^2 + x^2 = a^2 + a^2\tan^2\theta = a^2(1 + \tan^2\theta) = a^2\sec^2\theta.$$

With $x = a\sin\theta$,

$$a^2 - x^2 = a^2 - a^2\sin^2\theta = a^2(1 - \sin^2\theta) = a^2\cos^2\theta.$$

With $x = a\sec\theta$,

$$x^2 - a^2 = a^2\sec^2\theta - a^2 = a^2(\sec^2\theta - 1) = a^2\tan^2\theta.$$

EXERCISES 8.5

Expanding Quotients into Partial Fractions

Expand the quotients in Exercises 1–8 by partial fractions.

1. $\dfrac{5x - 13}{(x - 3)(x - 2)}$
2. $\dfrac{5x - 7}{x^2 - 3x + 2}$
3. $\dfrac{x + 4}{(x + 1)^2}$
4. $\dfrac{2x + 2}{x^2 - 2x + 1}$
5. $\dfrac{z + 1}{z^2(z - 1)}$
6. $\dfrac{z}{z^3 - z^2 - 6z}$
7. $\dfrac{t^2 + 8}{t^2 - 5t + 6}$
8. $\dfrac{t^4 + 9}{t^4 + 9t^2}$

Nonrepeated Linear Factors

In Exercises 9–16, express the integrand as a sum of partial fractions and evaluate the integrals.

9. $\displaystyle\int \frac{dx}{1 - x^2}$
10. $\displaystyle\int \frac{dx}{x^2 + 2x}$
11. $\displaystyle\int \frac{x + 4}{x^2 + 5x - 6}\,dx$
12. $\displaystyle\int \frac{2x + 1}{x^2 - 7x + 12}\,dx$
13. $\displaystyle\int_4^8 \frac{y\,dy}{y^2 - 2y - 3}$
14. $\displaystyle\int_{1/2}^1 \frac{y + 4}{y^2 + y}\,dy$
15. $\displaystyle\int \frac{dt}{t^3 + t^2 - 2t}$
16. $\displaystyle\int \frac{x + 3}{2x^3 - 8x}\,dx$

Repeated Linear Factors

In Exercises 17–20, express the integrand as a sum of partial fractions and evaluate the integrals.

17. $\displaystyle\int_0^1 \frac{3t}{x^2 + 2x + 1}$
18. $\displaystyle\int_{-1}^0 \frac{x^3\,dx}{x^2 - 2x + 1}$
19. $\displaystyle\int \frac{dx}{(x^2 - 1)^2}$
20. $\displaystyle\int \frac{x^2\,dx}{(x - 1)(x^2 + 2x + 1)}$

Irreducible Quadratic Factors

In Exercises 21–32, express the integrand as a sum of partial fractions and evaluate the integrals.

21. $\displaystyle\int_0^1 \frac{dx}{(x + 1)(x^2 + 1)}$
22. $\displaystyle\int_1^{\sqrt{3}} \frac{3t^2 + t + 4}{t^3 + t}\,dt$
23. $\displaystyle\int \frac{y^2 + 2y + 1}{(y^2 + 1)^2}\,dy$
24. $\displaystyle\int \frac{8x^2 + 8x + 2}{(4x^2 + 1)^2}\,dx$
25. $\displaystyle\int \frac{2s + 2}{(s^2 + 1)(s - 1)^3}\,ds$
26. $\displaystyle\int \frac{s^4 + 81}{s(s^2 + 9)^2}\,ds$
27. $\displaystyle\int \frac{x^2 - x + 2}{x^3 - 1}\,dx$
28. $\displaystyle\int \frac{1}{x^4 + x}\,dx$
29. $\displaystyle\int \frac{x^2}{x^4 - 1}\,dx$
30. $\displaystyle\int \frac{x^2 + x}{x^4 - 3x^2 - 4}\,dx$
31. $\displaystyle\int \frac{2\theta^3 + 5\theta^2 + 8\theta + 4}{(\theta^2 + 2\theta + 2)^2}\,d\theta$
32. $\displaystyle\int \frac{\theta^4 - 4\theta^3 + 2\theta^2 - 3\theta + 1}{(\theta^2 + 1)^3}\,d\theta$

7.8 EXERCISES

1. Explain why each of the following integrals is improper.

(a) $\int_1^\infty x^4 e^{-x^4}\,dx$ (b) $\int_0^{\pi/2} \sec x\,dx$

(c) $\int_0^2 \frac{x}{x^2 - 5x + 6}\,dx$ (d) $\int_{-\infty}^0 \frac{1}{x^2 + 5}\,dx$

2. Which of the following integrals are improper? Why?

(a) $\int_1^2 \frac{1}{2x - 1}\,dx$ (b) $\int_0^1 \frac{1}{2x - 1}\,dx$

(c) $\int_{-\infty}^\infty \frac{\sin x}{1 + x^2}\,dx$ (d) $\int_1^2 \ln(x - 1)\,dx$

3. Find the area under the curve $y = 1/x^3$ from $x = 1$ to $x = t$ and evaluate it for $t = 10$, 100, and 1000. Then find the total area under this curve for $x \geqslant 1$.

4. (a) Graph the functions $f(x) = 1/x^{1.1}$ and $g(x) = 1/x^{0.9}$ in the viewing rectangles $[0, 10]$ by $[0, 1]$ and $[0, 100]$ by $[0, 1]$.
(b) Find the areas under the graphs of f and g from $x = 1$ to $x = t$ and evaluate for $t = 10$, 100, 10^4, 10^6, 10^{10}, and 10^{20}.
(c) Find the total area under each curve for $x \geqslant 1$, if it exists.

7. $\int_{-\infty}^{-1} \frac{1}{\sqrt{2 - w}}\,dw$

8. $\int_0^\infty \frac{x}{(x^2 + 2)^2}\,dx$

9. $\int_4^\infty e^{-y/2}\,dy$

10. $\int_{-\infty}^{-1} e^{-2t}\,dt$

11. $\int_{-\infty}^\infty \frac{x}{1 + x^2}\,dx$

12. $\int_{-\infty}^\infty (2 - v^4)\,dv$

13. $\int_{-\infty}^\infty xe^{-x^2}\,dx$

14. $\int_1^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx$

15. $\int_{2\pi}^\infty \sin\theta\,d\theta$

16. $\int_{-\infty}^\infty \cos \pi t\,dt$

17. $\int_1^\infty \frac{x + 1}{x^2 + 2x}\,dx$

18. $\int_0^\infty \frac{dz}{z^2 + 3z + 2}$

19. $\int_0^\infty se^{-5s}\,ds$

20. $\int_{-\infty}^6 re^{r/3}\,dr$

21. $\int_1^\infty \frac{\ln x}{x}\,dx$

22. $\int_{-\infty}^\infty x^3 e^{-x^4}\,dx$

23. $\int_{-\infty}^\infty \frac{x^2}{9 + x^6}\,dx$

24. $\int_0^\infty \frac{e^x}{e^{2x} + 3}\,dx$

EXERCISES 8.8

Evaluating Improper Integrals

The integrals in Exercises 1–34 converge. Evaluate the integrals without using tables.

1. $\int_0^\infty \frac{dx}{x^2+1}$
2. $\int_1^\infty \frac{dx}{x^{1.001}}$
3. $\int_0^1 \frac{dx}{\sqrt{x}}$
4. $\int_0^4 \frac{dx}{\sqrt{4-x}}$
5. $\int_{-1}^1 \frac{dx}{x^{2/3}}$
6. $\int_{-8}^1 \frac{dx}{x^{1/3}}$
7. $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$
8. $\int_0^1 \frac{dr}{r^{0.999}}$
9. $\int_{-\infty}^{-2} \frac{2\,dx}{x^2-1}$
10. $\int_{-\infty}^{2} \frac{2\,dx}{x^2+4}$
11. $\int_2^\infty \frac{2}{v^2-v}\,dv$
12. $\int_2^\infty \frac{2\,dt}{t^2-1}$
13. $\int_{-\infty}^\infty \frac{2x\,dx}{(x^2+1)^2}$
14. $\int_{-\infty}^\infty \frac{x\,dx}{(x^2+4)^{3/2}}$
15. $\int_0^1 \frac{\theta+1}{\sqrt{\theta^2+2\theta}}\,d\theta$
16. $\int_0^2 \frac{s+1}{\sqrt{4-s^2}}\,ds$
35. $\int_{1/2}^2 \frac{dx}{x \ln x}$
36. $\int_{-1}^1 \frac{d\theta}{\theta^2-2\theta}$
37. $\int_{1/2}^\infty \frac{dx}{x(\ln x)^3}$
38. $\int_0^\infty \frac{d\theta}{\theta^2-1}$
39. $\int_0^{\pi/2} \tan\theta\,d\theta$
40. $\int_0^{\pi/2} \cot\theta\,d\theta$
41. $\int_0^1 \frac{\ln x}{x^2}\,dx$
42. $\int_1^2 \frac{dx}{x \ln x}$
43. $\int_0^{\ln 2} x^{-2}e^{-1/x}\,dx$
44. $\int_0^1 \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx$
45. $\int_0^\pi \frac{dt}{\sqrt{t}+\sin t}$
46. $\int_0^1 \frac{dt}{t-\sin t}$ (*Hint:* $t \geq \sin t$ for $t \geq 0$)
47. $\int_0^2 \frac{dx}{1-x^2}$
48. $\int_0^2 \frac{dx}{1-x}$
49. $\int_{-1}^1 \ln|x|\,dx$
50. $\int_{-1}^1 -x \ln|x|\,dx$
51. $\int_1^\infty \frac{dx}{x^3+1}$
52. $\int_4^\infty \frac{dx}{\sqrt{x}-1}$